



AP Calculus Practice Set Answers

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Range: AP Calc 1.1-10.3, Pre-calc Chapters 9-12

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

I. Calculus

1. a) 2

b) 0

c) DNE

d) 3

e) -2

f) 0

2. a) $2x+1$

b) $8x^3 - 23$

c) $2 \cos(2x)$

d) $2\cos(x)\cos(2x) - \sin(x)\sin(2x)$

e) $2^x \ln 2$

f) $-4\cos(x^3 + \sin(2x))\sin(x^3 + \sin(2x))(3x^2 + \cos(2x))$

g) $-\frac{\sec^2(x)}{\sqrt{1 - \frac{1}{\tan^2(x)}} \tan^2(x)}$

h) $\ln(3)$

3. Evaluations

a) $1/3$

b) 10

c) DNE

d) $e - 1$

e) 1

f) $-56/3$

g) DNE

h) $\pi/2$

4. Area and Volume

a) $12+20\pi-40\tan^{-1}(3)=24.87$

b) $3\pi/10$

c) $\pi/105$

5. Polar Curve

a) 6π

b) $\pi/2$

c) r negative $\Rightarrow$ point is reflected over origin

d) $\theta = \pi/2$

6. Taylor Series

a) $f'''(0) = -2$

b) $(-1)^{n-1} / n$

c) $(1)(0.5)^4/24 < \frac{1}{100}$

d) $\frac{x^5}{3} - \frac{x^4}{2} - x^3$

II. Pre-calculus

1. a) $\pi/6$

b) $2\pi/3$

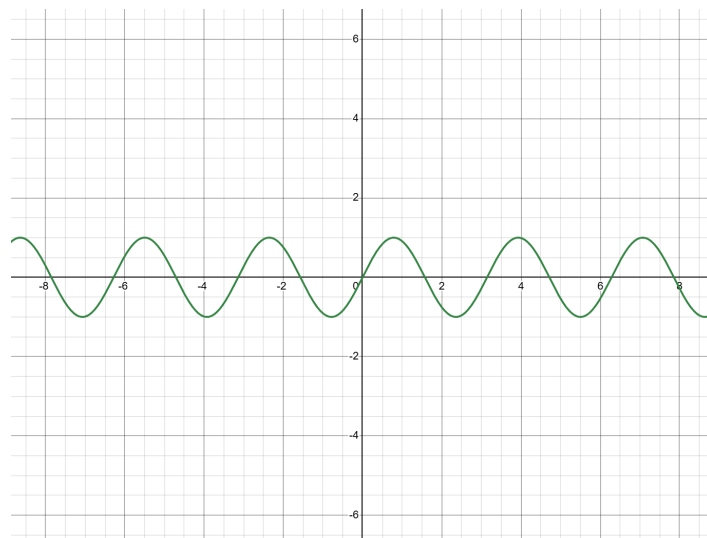
c) $\pi/4$

d) $\pi/6$

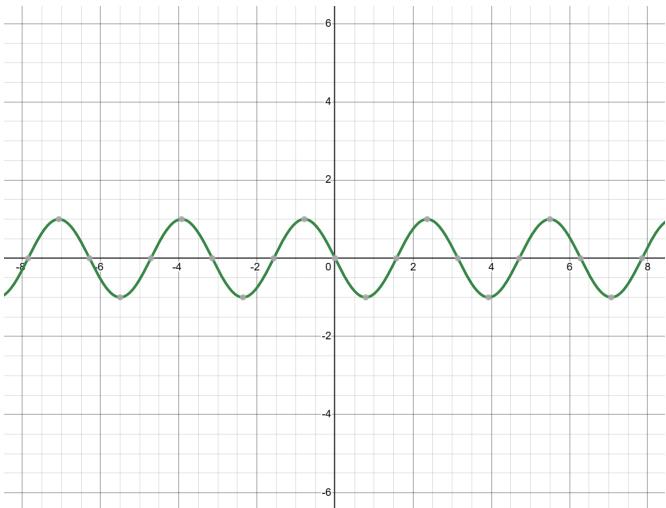
e) $\pi/4$

f) 0

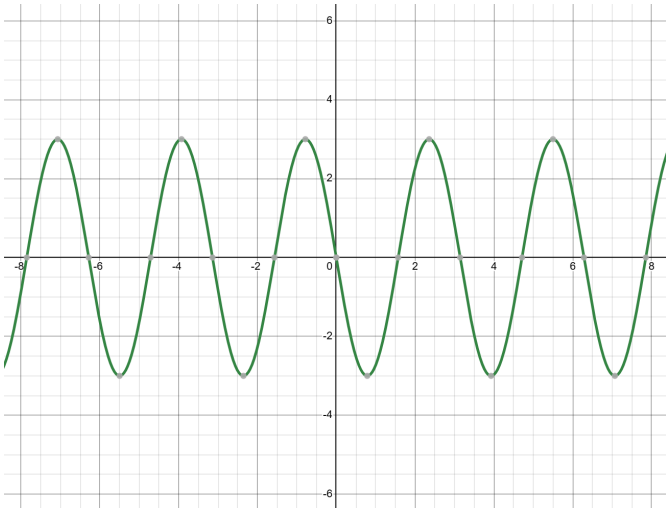
2. a)



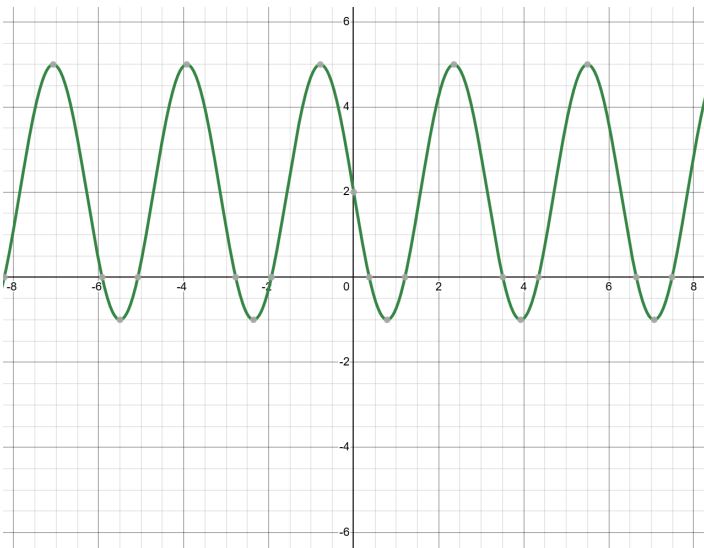
b)



c)



d)



3. $r = 6$

4. $\theta = 240^\circ$

5. $\theta = 62.49^\circ$

6. 6

7. $a = -1, b = -5$